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XGBoost To Enhance Learner Performance Prediction

Soukaina Hakkal *, Ayoub Ait Lahcen

Engineering Sciences Laboratory, National School of Applied Sciences, Ibn Tofail University, Kenitra, Morocco

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ABSTRACT

The huge amount of data generated by an Intelligent Tutoring System becomes useful when analyzed in an appropriate way to provide significant insights about learners, especially his or her performance. Performance data retrieved from historical interactions is the main engine for learner performance prediction, where the likelihood of the learner answering correctly future questions is calculated. Modeling learner performance can provide significant insights into individual students to promote successful learning and maximize educational achievement. This study aims to enhance the learner performance prediction of some logistic regression-based models, namely Item Response Theory, Performance Factor Analysis, and DAS3H using XGBoost, including an empirical comparison of eight real-world datasets, containing performance log data collected from different online intelligent tutoring systems, involving the first time a new dataset from Moodle Morocco. The results have demonstrated that the XGBoost has enhanced PFA predictive performance on seven datasets with an AUC of up 0.88 and improved the DAS3H AUC on the ASSISTment17 dataset while conserving almost the same predictive results for Item Response Theory on some datasets.

1. Introduction

Computer-assisted instruction (CAI), also known as computer-assisted learning (CAL), emerged in the 1950s with the development of 'Linear Programs' by Skinner (Atkinson & Wilson, 1968). These early CAI systems were text-based focused on providing programmed instructions without including the necessary feedback or individualization.

The concept of Intelligent Tutoring Systems (ITS), pioneered by (Anderson et al. ,1985), evolved from generative CAI systems (Nwana, 1990). Unlike early CAI, ITS involves customized learning materials that adapt to learners' abilities and learning pathways. For an effective ITS, it must incorporate robust knowledge assessment to understand learner's current knowledge state and track their learning progress (Wang, 2007). This ongoing assessment process helps determine learners' comprehension and serves as the foundation for adaptating instructional materials. Immediate feedback during the learning process is crucial (Shute, 2008; Taghizadeh Kerman et al., 2024), as is the recommendation of relevant learning content based on individual needs (Hsu et al., 2010).

In the era of Big Data, educational systems are generating massive amounts of information on learners' interactions. This data provides an opportunity to analyze learner behavior in detail, gaining insights into their knowledge levels and learning preferences, thereby enhancing their educational experience. However, assessing a student's knowledge of a specific skill is challenging. To address this, ITS integrates a domain model (Kass, 1989) to extract knowledge components (KCs) from the learner's data. This domain model offers an active representation of the learner's evolving knowledge and skills. KCs derived from the domain model are instrumental in assessing a student's understanding and proficiency within the student model of ITS. This a knowledge assessment information contributes to modeling learner performance (Hakkal & Ait Lahcen, 2021).

Modeling and predicting learner performance involves analyzing a learner's interactions within an ITS to estimate the probability that they will answer certain questions correctly, based on underlying KC in the system (Kosmas & Zaphiris, 2023). Accurate predictions of academic performance offer significant benefits for teachers, students, policy-makers, and institutions (Rodríguez-Hernández et al., 2021). These predictions can also enhance learning outcomes and increase goal attainment within adaptive learning environments (Matzavela & Alepis, 2021).

One of the classic methods for tracking student knowledge and predicting future success is Bayesian Knowledge Tracing (BKT) (Corbett & Anderson, 1995; Liu, 2022). BKT is based on a Hidden Markov Model (HMM), where a skill is a binary latent feature either learned or

* Corresponding author.
E-mail addresses: soukaina.hakkal@uit.ac.ma (S. Hakkal), ayoub.aitlahcen@uit.ac.ma (A.A. Lahcen).

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unlearned. The model assumes that once a skill is learned, the learner does not forget it, and each action applies to only one skill. The overall likelihood in BKT is calculated using four main parameters: initial probability, transition probability, guess probability, and slip probability. Deep Knowledge Tracing (DKT) was developed by applying deep learning techniques, such as Recurrent Neural Networks (RNN), to the classical HMM framework (Piech et al., 2015). The DKT architecture has two key components: a knowledge state model that tracks a learner's progress over time, and a performance model that predicts a learner's performance on future questions.

In addition to BKT and DKT, logistic regression-based models have also proven effective in predicting student performance. A simple logistic regression-based model is the Item Response Theory (IRT) (Wilson & De Boeck, 2004), which predicts a student's future performance based on item difficulty alone. Performance Factor Analysis (PFA) is another approach, using counts of correct and incorrect responses to track past practice, without considering item difficulty (Pavlik Jr et al., 2009). However, DAS3H combines elements of IRT and PFA, adding a time-window feature to account for learning and forgetting while predicting student performance (Choffin et al., 2019).

eXtreme Gradient Boosting (XGBoost) is a scalable tree-boosting algorithm designed for high performance, adaptability, and mobility, delivering state-of-the-art results across a variety of data science applications. It excels at handling sparse data efficiently (Chen & Guestrin, 2016). Despite its strengths, XGBoost has rarely been applied to predict learner performance, and it has never been combined with logistic regression-based models such as IRT, PFA, and DAS3H. Investigating these combinations could potentially enhance their predictive capabilities and push computational limits in educational contexts.

This paper introduces two main contributions within the educational data mining (EDM) field, we seek to present how logistic regression-based models such as IRT, PFA, and DAS3H could be further enhanced using the XGBoost model, producing reliable predictions of students' aptitude for handling future question. We perform an empirical study to compare Logistic regression with XGBoost using log data of students taking different courses, from eight real-world datasets involving for the first time a new dataset from Moodle Morocco.

This paper makes two significant contributions to the field of EDM. First, it explores how logistic regression-based models, specifically IRT, PFA, and DAS3H, can be further enhanced with the XGBoost model to produce more reliable predictions of students' aptitude in handling future questions. Second, it presents an empirical study that compares the performance of logistic regression with XGBoost, using log data from students enrolled in various courses. This study uses eight real-world datasets, including, for the first time, a new dataset from Moodle Morocco.

XGBoost has outperformed state-of-the-art results in predictive performance on seven datasets for the PFA model, showing significant improvements in the AUC metric. It also surpassed logistic regression on the ASSISTment17 dataset for DAS3H, where the AUC increased from 0.690 to 0.709. For other models and datasets, the XGBoost results closely align with those of logistic regression, confirming its reliability.

This paper is structured as follows: Section 2 provides a global overview of relevant research in modeling learner performance. In Sections 3 and 4, we discuss the various logistic regression-based models and the XGBoost model, respectively. Section 5 describes the research methodology, including data collection and experiment setting. including the results and discussion, and the final section summarizes our key findings and conclusions.

2. Related work

The purpose of this research is to improve the learner performance prediction of previous logistic regression models using the boosting machine learning technique, and to provide an empirical comparison between state-of-the-art evaluation metrics and boosting techniques.

Several studies have investigated different machine learning models for predicting learner performance. In the study of (Alsulami et al., 2023), the authors analyzed and predicted learner performance using log data from an Intelligent Tutoring System during the COVID-19 pandemic. They used three machine learning models; Random Forest, Naive Bayes, and Decision Tree, and improved results through an ensemble technique including bagging, boosting, and voting. Overall accuracy increased from 0.75 to 0.77 with the addition of boosting. (Asselman et al., 2023) proposed a new PFA technique based on Random Forest, AdaBoost, and XGBoost to improve learner performance prediction. The results showed that XGBoost outperformed the other models, significantly enhancing predictive accuracy. (Fan et al., 2023) proposed a dual particle swarm optimization technique called P2catBoost for predicting secondary school performance. The P2catBoost model outperformed other models and demonstrated its effectiveness across various datasets.

In (Yan, 2021) study, XGBoost was used to predict learner performance using a new macro-educational structured dataset. The authors compared XGBoost's results with those of Random Forest, Lasso, Elastic Net, Support Vector Machine, and Decision Tree, with XGBoost demonstrating the best predictive results. (Ojajuni et al., 2021) employed a range of machine learning models including Random Forest, Support Vector Machine, Gradient Boosting, Decision Tree, Logistic Regression, Deep Learning, and XGBoost to predict learner performance. The results showed that XGBoost was highly effective at forecasting learner performance. (Nahar et al., 2021) examined the performance of engineering school students and used six classification algorithms to predict their results. The comparison, which included considerations of predictive accuracy, complexity, and other factors, found that Support Vector Machine had the best overall performance based on sensitivity analysis of the datasets. In a comparative study by (Y. Chen & Zhai, 2023), the authors used various machine learning algorithms to predict student performance. The results indicated that Random Forest achieved higher generality, while Decision Tree and artificial neural networks demonstrated strong abilities to predict learner performance.

In this research by (Gervet et al., 2020), the authors conducted a comprehensive empirical comparison across nine real-world datasets to determine which model offers better prediction accuracy under various conditions. They used Logistic Regression, DKT, and BKT to predict learner performance. Their analysis included ablation studies and model calibration to ensure reliability. The results showed that Logistic Regression, with a proper selection of features, achieved the best performance, followed by DKT on large datasets. Another study by (Schmucker et al., 2021) aimed to improve the predictive performance of logistic regression models by introducing new features derived from response logs to conventional questions. This study also incorporated additional features related to video sequences and contextual information. Their approach involved combining several specialized models to create a prediction cluster for learner performance. This approach yielded a significant improvement in the Area Under the Curve (AUC), from 0.767 to 0.808, across four new datasets.

3. Logistic regression for learner performance prediction

Logistic regression is a commonly used technique in EDM due to its reliable ability to predict learner performance. Consider a student's interactions within an intelligent tutoring system (ITS) as $x_t = (x_1, \ldots, x_t)$ where t^{th} is a tuple of $x_t = (q_t, a_t)$ q_t is the ID question that the student attempted to answer and a_t indicates whether the student's answer is correct (binary value) (Schmucker et al., 2021).

The performance prediction involves estimating the probability that a student will correctly answer the next question given as $p(a_{t+1} = 1 | q_{t+1}, x_{1:t})$. Therefore, each interaction may involve one or more knowledge components (KCs) for a given question KC(q_t). For models that consider learning and forgetting, the timestamp feature (TS_t) is included to track

temporal dynamics in the learning process.

Giving that.

- s learner;
- k knowledge component (KC);
- i item;
- TS_t Timestamp at time t;
- q_t question at time t;
- $a_t \in \{0, 1\}$ correctness at time t;
- $x_{1:t}$ all data up to time t (interactions);
- $x_{s,1:t}$ all data of learner s up to time t (learner interactions);
- $\delta(.)$ one hot encoding of feature.

The logistic regression prediction of the student (s) answering the question $\left(q_{s,t+1}\right)$ correctly $\left(a_{s,t+1}\right)$ at time t, is represented mathematically as follows:

$$p\left(a_{s,t+1}=1|q_{s,t+1}, x_{s,1:t}\right)=\sigma\left(w^T\Phi\left(q_{s,t+1}, x_{s,1:t}\right)\right)$$

Giving that.

- σ logistic function identified as $\sigma(x) = {}^{1}/_{(1+e^{-x})}$, represents the sigmoid function that generates a value ranging from 0 to 1, expressed as the likelihood that student s will correctly answer question q_{t+1};
- w^T a trainable weight vector;
- $\Phi = (\varphi 1, \ldots, \varphi d)$ a features vector.

Given that a variety of Φ feature vectors have led to various models in the literature, the first one is Item Response Theory (IRT) (W. J. van der Linden & R. K. Hambleton, 2013). IRT is a probabilistic model built to simulate the learner's ability αs which is a latent trait estimated based on the learner's responses to a set of questions, and $\delta q_{s,t+1}$ the question difficulty which is quantified from an analysis of the learners' responses to the item, and the ability level where the likelihood of correct response is 0.50, aiming to estimate the likelihood of student s to correctly solve a question qs,t+1 as follows:

$$p_{IRT}\left(a_{s,t+1}=1|q_{s,t+1}, x_{s,1:t}\right)=\sigma\left(\alpha_s-\delta q_{s,t+1}\right)$$

Item Response Theory (IRT) is based on the premise that a learner's ability to predict their performance directly influences their capacity to answer questions correctly. In other words, learners with higher skill levels are more likely to get the correct answer compared to those with lower skill levels. According to the IRT model, a student's level remains constant and does not change with practice (Reckase, 2009). The relationship between the likelihood of a correct answer and the learner's ability is visualized through an Item Characteristic Curve (ICC) (Fig. 1). This curve illustrates how the probability of answering correctly increases with the learner's ability (An & Yung, 2014).

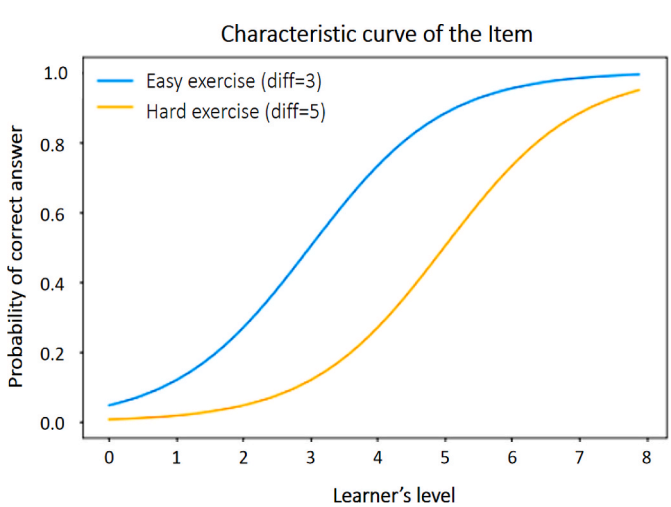

Fig. 1. Item characteristic curve.

The second feature vector, PFA, derived from the AFM, uses win/fail counters to track historical successful or unsuccessful attempts for each KC (Pavlik Jr et al., 2009; Cen et al., 2006). The mathematical representation of the model is give:

$$p_{PFA}\left(a_{s,t+1}=1|q_{s,t+1}, x_{s,1:t}\right)=\sigma\left(\sum_{k\in KC\left(q_{s,t+1}\right)}\beta_k+\beta\gamma_k c_{s,k}+\rho_k f_{s,k}\right)$$

Giving that.

- β_k difficulty of KC k;
- $c_{s,k}$ counter of right responses of student s on KC k before this attempt;
- $f_{s,k}$ counter of the incorrect response of students s on KC k before this attempt.

PFA capable of handling multi-skill questions, deliberately avoids considering item difficulty to prevent future item cold start issues (Vie & Kashima, 2019). According to earlier studies, PFA was as predictive as DKT (X. Xiong et al., 2016) and outperformed the KT model (Gong et al., 2010).

The third feature vector, DAS3H (Choffin et al., 2019), represents the combination of four factors: item Difficulty, student Ability, Skill, and Student Skill practice History. This model is predictive and interpretable, allowing for a nuanced understanding of KCs, it is used to examine both the learning and forgetting of KCs. Based on the DASH model (Lindsey et al., 2014), DAS3H incorporates forgetting and practice effects by accounting for the structured timing of past experience with a given KC. This approach enables the model to represent evolving learning and forgetting curves for each KC.

DAS3H combines elements from both PFA and IRT to predict whether a learner s responds correctly $\left(a_{s,t+1}\right)$ to an item $\left(q_{s,t+1}\right)$ linked to 1 or more KC k. This model introduces counters of the learner s's correct responses on a KC within a given w, as well as counters of attempts for learner s on the same KC k in a time window w. These counters allow to capture both the learning and forgetting of KCs from the learner's result history as follows:

$$p_{DAS3H}\left(a_{s,t+1}=1|q_{s,t+1}, x_{s,1:t}\right)=\sigma\left(\alpha_s-\delta_{q_{s,t+1}}+\sum_{k\in KC\left(q_{s,t+1}\right)}\beta_k + \sum_{k\in KC\left(q_{s,t+1}\right)}\sum_{w=0}^{W-1}\theta_{k,2w+1}\varnothing\left(c_{s,k,w}\right)-\theta_{k,2w+2}\varnothing\left(a_{s,k,w}\right)\right)$$

Giving that.

- $w \in \{0, W-1\}$ a series of growing time windows;
- $c_{s,k,w}$ right response counter of student s on KC k in time window w;
- $a_{s,k,w}$ attempts counter of student s on KC k in time window w;
- $\varnothing(x) = log(1+x)$ revalues the counters.
- $\theta \in \mathbb{R}^{k\times 2W}$ the matrix of coefficients related to the success $c_{s,k,w}$ and attempt $a_{s,k,w}$ counters. These coefficients are used to describe the learning and forgetting curve of a kC k.

In most learning algorithms, there's a focus on either considering the phenomenon of forgetting or establishing the relationship between items and KC, but rarely both simultaneously. For instance, (Ekanadham & Karklin, 2017) incorporate temporality into the IRT model without

explicitly addressing forgetting, and they don't allow for an item to be linked to multiple KCs at once. Similarly, traditional BKT focuses on the sequence of interactions while ignoring elapsed time. Although DKT addresses human forgetting by considering the time between interactions and the counter of previous attempts, it can't model items with multiple KCs directly (Nagatani et al., 2019).

Given these limitations, the DAS3H model fills the gap by accounting for both dimensions: it tracks correct responses and attempts over time to measure forgetting and learning, and it supports items with multiple KCs. This design is specifically tailored to help learners master a set of KCs through consistent practice, acknowledging that each item might involve more than one KC.

The experiments by Choffin reveal that integrating time windows into the model can significantly enhance performance compared to KTM, while unique encoding for each KC also contributes to improved results (Lai et al., 2021). This dual approach, combining time-based elements and multiple KCs, provides a comprehensive framework for understanding learning and forgetting (see Table 1).

4. Extreme gradient boosting

XGBoost (T. Chen & Guestrin, 2016), is an optimized distributed framework for gradient boosting designed for high effectiveness, adaptability, and portability. Its goal is to leverage the extreme computational capabilities of modern hardware to offer a scalable and accurate end-to-end boosting system. As an enhanced version of Gradient Boosting Regression Trees (GBRT) (Friedman, 2001), XGBoost introduces parallel tree boosting to address various data science challenges with precision, speed, and efficiency. It achieves this through several approximations and practical techniques that dramatically accelerate the learning process. Friedman's work laid the groundwork for a range of boosting algorithms, including AdaBoost (Freund et al., 1999), Gradient Boosting (Friedman, 2001), LightGBM (Ke et al., 2017), and XGBoost. The fundamental principle of boosting is the parallel combination of weak learners, typically decision trees, to construct a strong learner. The approach involves fitting new models to the residuals of the previous models and iteratively minimizing the loss function to refine the accuracy of the predictions. Through this iterative process, boosting techniques incrementally improve performance, transforming weak learners into a robust ensemble model.

The objective function being optimized by XGBoost for regression case, take the form:

$$\Theta = \sum_{i=1}^{M} L(y_i, \widehat{y}_i) + \sum_{j=1}^{J} \Omega\left(f_j\right)$$

Where.

- M samples number in the dataset;
- y_i true value for ith sample;
- $\widehat{y}_i$ model-predicted value;
- L Loss function, the average squared difference between the actual and the predicted value, referred to as the mean squared error (MSE);
- $\Omega\left(f_j\right)$ regularization parameter to avoid overfitting by penalizing tree complexity while adding a penalty term to the loss function;
- f_j the jth tree in the set.

Finally, the XGBoost prediction $\widehat{y}_i$ can be expressed as follows:

$$\widehat{y}_i = \sum_{j=1}^{J} f_j(x_i)$$

Where x_i designs the input parameters required for the ith sample, and $f_j(x_i)$ designs the prediction of the jth tree of the ith sample.

For classification, the objective function using log loss or binary entropy is expressed as [37]:

$$\Theta = \sum_{i=1}^{M} [y_i \log(1 + e^{-\hat{y}_i})(1 + y_i)\log\,(1 + e^{\hat{y}_i})] + \sum_{j=1}^{J} \Omega\left(f_j\right)$$

Which can be represented as the sum of the logistic loss component and the regularization term, where.

- M samples number in the dataset;
- y_i true class label for ith sample;
- $\widehat{y}_i$ positive class predicted probability for ith sample;
- $\Omega\left(f_j\right)$ regularization parameter.
- $y_i \log(1 + e^{-\hat{y}_i})$: when the true label y_i is 1, in this case, the model is penalized whenever class 0 (negative) is predicted with a high degree of confidence when $\widehat{y}_i$ is large and negative, where $\log(1 + e^{-\hat{y}_i})$ rises as class 0's predicted probability $e^{-\hat{y}_i}$ drops, further penalizing the model for misclassification.
- $(1 + y_i)\log\,(1 + e^{\hat{y}_i})$: when the true label y_i is 0, in this case, the model is penalized whenever class 1 (positive) is predicted with a high degree of confidence when $\widehat{y}_i$ is large and positive, where $\log\,(1 + e^{\hat{y}_i})$ linearly rises with $\widehat{y}_i$, penalizing the model more for classification errors in class 1.

Hence, the XGBoost predictive probability takes the following form:

$$p(y = 1|x_i) = \frac{1}{1 + e^{-\sum_{j=1}^{J} f_j(x_i)}}$$

Where x_i designs the input parameters required for the ith sample, and $f_j(x_i)$ designs the prediction of the jth tree of the ith sample.

The XGBoost engine starts by defining an initial prediction value that represents the average of the target values. This is followed by calculating the initial residual, which is the difference between the target value and the predicted value. The boosting iteration then begins. In each round, a new tree is fitted based on the previous residual, allowing the model to improve with each iteration. This iterative approach is driven by the loss gradient, with each new tree added in proportion to the learning rate (also known as shrinkage).

Regularization techniques are applied to control model complexity and prevent overfitting. Finally, optimization methods are used to minimize the objective function as discussed by (T. Chen & Guestrin, 2016). This iterative process allows the model to refine its predictions over time, leading to improved accuracy.

5. Methodology

In this section, we introduce the experiment settings by describing the datasets selected to perform this study, then the experiment methodology and the evaluation metrics.

Table 1
Summary of the main features of the IRT, PFA, and DS3H models (Sein Minn, 2022).

Φ	Data type	Skill's type	Learner ability	Interpretability	Sequentially	Learning transfer	Item information	Predictive performance
IRT	Static & Dynamic	Single	Continuous	Yes	No	Yes	Yes	High
PFA	Dynamic	Multiple	Continuous	Yes	No	Yes	No	Medium
DAS3H	Dynamic	Multiple	Continuous	Yes	Yes	Yes	Yes	High

5.1. Datasets

This study was based on eight commonly used real-world datasets.

From the ASSISTments Intelligent Tutoring System, four skill builder datasets were used: ASSISTments 2009–2010 (***assist09***), ASSISTments 2012–2013 (***assist12***), ASSISTments 2015 (***assist15***), ASSISTments Challenge 2017 (***assist17***) (Feng et al., 2009; Gervet et al., 2020). Where ASSISTments is an online intelligent tutoring system for mathematics, offering qualitative and quantitative assessment, in conjunction with student support and feedback, as well as teachers' comprehensive reports.

From the Knowledge Discovery and Data Mining, KDD Cup Challenge 2010 two datasets were used: Algebra 2005–2006 (***algebra05***), and Bridge Algebra 2006–2007 (***bridge-algebra06***) (Stamper et al., 2010). Where KDD is a data mining challenge that aims to provide a real-world dataset to extract useful insight and develop advanced algorithms to address challenging data analysis tasks.

From the PSLC Datashop, the Statics dataset is used, where the log data are collected from college-level engineering ***Statics*** courses that focus on mathematical methods used to solve engineering problems related to forces and moments (Koedinger et al., 2010).

From Moodle, an open-source learning management system, that is extensively integrated into educational establishments to support course management and evaluation, the ***Moodle-morocco*** dataset was selected. Where performance log data was recorded and collected from initiation to the programming language (python language) course from the National School of Applied Science in Morocco (Arkiza et al., 2023). The dataset categorizes multiple-choice questions into distinct skills. This classification helps to explore student performance across each skill individually, thereby allowing a comprehensive understanding of student areas of strength and weakness in different aspects of programming.

We note a major variation between datasets in terms of the number of KC (the specific elements of knowledge or concepts required for a particular domain (skills)) covered, items, number of learners, and temporal characteristic (timestamps), which is not available on all the datasets, and only used by the DAS3H model. We have observed that the volume of the dataset has directly impacted the overall performance of the models, especially the XGBoost model. The larger the dataset size, the more performance is enhanced, due to the ability of the XGBoost model to capture a wider range of patterns and relationships to help the model generalize better and make accurate predictions. Consequently, where generalization is concerned, the overall volume of a dataset is influenced not only by the actual amount of students and the actual amount of interactions but, more significantly, by the number of learners attempting each item and each knowledge component.

The datasets exploited in tutoring systems have different characteristics that have an impact on the performance of prediction models. The level of sequential progression of students through the material affects model performance. Most data sets follow a consecutive order, where consecutive item numbers correspond to practice questions on the same topic (Table 2.).

Table 2
Datasets characteristics.

Dataset	Learners	Items	Skills	Timestamps
assist09	3241	17,709	124	No
assist12	29,018	53,086	265	Yes
assist15	14,567	100	100	No
assist17	1708	3162	102	Yes
bridge-algebra06	1146	129,263	493	Yes
algebra05	574	173,113	112	Yes
Statics	282	1223	98	No
Moodle-morocco	232	27	7	No

5.2. Experiment

5.2.1. Data preprocessing (data preparation, data encoding)

Preparing data for machine learning is a crucial step, especially since our students' log data can't be directly used in our models. The raw data must be cleaned and transformed to ensurqe that our models are not biased. This data preparation process involves various techniques and operations to improve quality, accuracy, and reliability, ultimately enhancing the effectiveness of the final model.

After collecting the required data, the raw data is thoroughly examined for its structure and features. This includes handling missing values, where any interactions with missing skill tags are removed, renaming columns, extracting timestamps, filtering continuous outcomes, filtering sequences based on a minimum number of interactions per user, and sorting data both temporally and by users. Similar preprocessing steps were applied to the ASSISTments dataset, the KDD dataset, and the Statics dataset. However, the Moodle-Morocco dataset required additional steps to adapt it to the context.

Following this, the Q-matrix is constructed as part of the data preprocessing. The Q-matrix plays a key role in many cognitive diagnostic models and educational data analysis procedures. It is a binary matrix representing item-skill relationships, where rows denote items (questions) and columns correspond to skills or knowledge components. The entries in this matrix indicate which skills are covered by a particular item. After identifying the skills and the items in the datasets, the Q-matrix is initialized as a matrix of zeros. The relevant entries are then assigned the value '1' to show that the item is linked to a specific skill. To efficiently store this binary relationship between items and skills, the Q-matrix is represented as a sparse array. This approach is beneficial when the number of skills is smaller than the total number of items, allowing memory optimization by storing only non-zero data and their positions.

The Q-matrix is a key element in the next step, which consists of building a sparse data set from this Q-matrix and the dense data set. Hence constructing a sparse dataset is based on different feature selections, including user one-hot encoding ('u'), item one-hot encoding ('i'), skills encoding ('s'), historical counts for items ('ic'), historical counts for skills ('sc'), total historical counts ('tc'), historical counts including wins ('w'), historical counts including attempts ('a'), and time window-based encoding of historical counts ('tw'). Where each feature selection is defined for a specific model, [-i] for item response theory IRT, [-s -sc -w -a] for PFA, and [-i -s -sc -w -a -tw] for DAS3H. Then a resulting sparse features matrix is saved as (.npz) to be used for the performance prediction.

5.2.2. Evaluation methodology

Following the train-test split and finalization of the Q-matrix, the XGBoost model is constructed using the XGBoost library in Python, which offers an optimized gradient-boosting algorithm. To validate the model's performance, we conducted a 5-fold cross-validation at the student level, as suggested by previous research. This approach entails dividing the learner population into 5 disjoint sets, rotating each set as a test set while utilizing the remaining 4 sets for training the model and selecting the optimal hyperparameters (Gervet et al., 2020).

The performance results are summarized for the best-selected hyperparameters. We employed 1000 trees, representing the number of iterations in the model. Increasing this number helps reduce error on the training set (bias), but excessively high values may lead to overfitting. A learning rate of 0.1 was chosen to enhance XGBoost's generalization ability, albeit at the cost of increased computational time during training and prediction. For reg-lambda, we opted for a value of 1.0 to control overfitting by penalizing high weights. This parameter plays a pivotal role in curbing model complexity, thereby improving generalization on unseen data, reducing variance, and enhancing model robustness. In comparing our results to prior studies, we selected AUC, RMSE, and ACC as evaluation metrics. AUC, representing the Area Under the ROC Curve, plots the true positive rate against the false

positive rate across different threshold values. RMSE, or Root Mean Square Error, quantifies the average error magnitude between predicted and observed values. Additionally, we considered ACC, which measures the overall correctness of predictions (Fig. 2).

We processed the data using code from the "learner-performance-prediction" repository on GitHub (Theo, 2020).

6. Results

In this section, we present the performance results for each dataset trained using the XGBoost model, we compare these results to those obtained using a state-of-the-art logistic regression model, as shown in Table 3 through Table 10. With area under the curve (AUC) and accuracy (ACC) shown for both models and datasets. The root mean square error (RMSE) is not shown for lack of space.

Our methodology outperforms state-of-the-art logistic regression on seven datasets in the case of Performance Facort Analysis (PFA), and on one dataset in the case of DAS3H. However when considering Item Response Theory (IRT), Logistic regression and XGBoost exhibit quite similar performance scores.

7. Discussion

In this study, we set out to improve the performance of Logistic Regression models through the integration of the booting model. Our research questions included evaluating the ability of the boosting technique through the XGBoost model to increase the predictive performance and strength of Logistic Regression over multiple real-world datasets. Taking forward the XGBoost power to overcome the limitations of state-of-the-art Logistic regression-based models, we sought to achieve a higher accuracy and more reliable solution.

In the following sections, the experimental results are discussed and analyzed, and our views presented on the effectiveness of the suggested methodology in accomplishing our objectives.

Our experimental results show that the XGBoost model with its best hyperparameter selection after the 5 cross-validations {iter:1000, reg-lambda:1.0, learning_rate:0.1} outperformed the logistic regression model with the “lbfgs” solver on seven datasets in the case of PFA. To confirm XGBoosts’ effectiveness in a variety of data contexts, the reported improvements in the Area Under the Roc curve are notable across all datasets. For bridge_algebra06, the AUC saw a remarkable increase of +0.025 (see Table 3), while +0.026 for algebra05 (see Table 4), +0.03 for assist17 (see Table 6), +0.023 for assist12 (see Table 5), and +0.013 for assist15 (see Table 8). While for the smallest datasets like Statics, the AUC has improved from 0.691 to 0.719 (see Table 9). therefore, the Moodle_Moroccodataset recorded the highest improvement of +0.042 (see Table 10).

In the context of IRT, our model achieves performance levels

Table 3
Algebra06 evaluation results.

Bridge-Algebra06	ACC	AUC
DAS3H-LR	0.849	0.790[1]
DAS3H-XGBOOST	0.846	0.783
PFA-LR	0.841	0.739[1]
PFA-XGBOOST	**0.843**	**0.764**
IRT-LR	0.839	0.747[1]
IRT-XGBOOST	0.837	0.639

Sources: [1] (Yeung & Yeung, 2018),[2] (Gervet et al., 2020),[3] (Arkiza et al., 2023).

Table 4
Algebra05 evaluation results.

Algebra05	ACC	AUC
DAS3H-LR	0.817	0.826[1]
DAS3H-XGBOOST	0.812	0.822
PFA-LR	0.788	0.744[1]
PFA-XGBOOST	**0.791**	**0.770**
IRT-LR	0.801	0.771[1]
IRT-XGBOOST	0.789	0.719

Sources: [1] (Yeung & Yeung, 2018),[2] (Gervet et al., 2020),[3] (Arkiza et al., 2023).

Table 5
Assist12 evaluation results.

Assistments12	ACC	AUC
DAS3H-LR	0.743	0.739[1]
DAS3H-XGBOOST	0.729	0.727
PFA-LR	0.714	0.670[1]
PFA-XGBOOST	**0.716**	**0.693**
IRT-LR	0.721	0.702[1]
IRT-XGBOOST	0.706	0.615

Sources: [1] (Yeung & Yeung, 2018),[2] (Gervet et al., 2020),[3] (Arkiza et al., 2023).

Table 6
Assist17 evaluation results.

Assistments17	ACC	AUC
DAS3H-LR	0.677	0.693[2]
DAS3H-XGBOOST	**0.685**	**0.709**
PFA-LR	0.647	0.620[2]
PFA-XGBOOST	**0.659**	**0.650**
IRT-LR	0.666	0.681[2]
IRT-XGBOOST	**0.666**	0.672

Sources: [1] (Yeung & Yeung, 2018),[2] (Gervet et al., 2020),[3] (Arkiza et al., 2023).

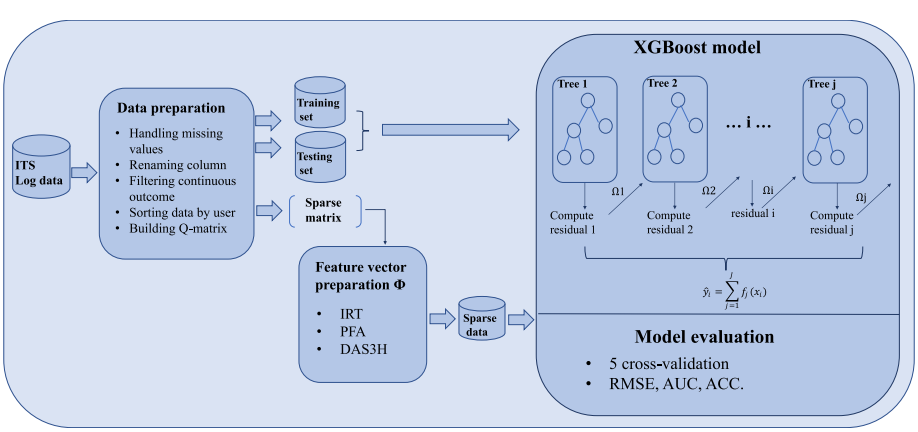

Fig. 2. XGBoost Learner performance prediction framework.

Table 7
Assist09 evaluation results.

Assistments09	ACC	AUC
PFA-LR	0.706	0.724[2]
PFA-XGBOOST	**0.708**	0.720
IRT-LR	0.684	0.692[2]
IRT-XGBOOST	0.681	0.633

Sources: [1] (Yeung & Yeung, 2018),[2] (Gervet et al., 2020),[3] (Arkiza et al., 2023).

Table 8
Assist15 evaluation results.

Assistments15	ACC	AUC
PFA-LR	0.739	0.690[2]
PFA-XGBOOST	**0.740**	**0.703**
IRT-LR	0.726	0.638[2]
IRT-XGBOOST	**0.726**	**0.644**

Sources: [1] (Yeung & Yeung, 2018),[2] (Gervet et al., 2020),[3] (Arkiza et al., 2023).

Table 9
Statics evaluation results.

Statics	ACC	AUC
PFA-LR	0.775	0.691[2]
PFA-XGBOOST	**0.780**	**0.719**
IRT-LR	0.805	0.789[2]
IRT-XGBOOST	0.804	0.794

Sources: [1] (Yeung & Yeung, 2018),[2] (Gervet et al., 2020),[3] (Arkiza et al., 2023).

Table 10
Moodle-Morocco evaluation results.

Moodle-Morocco	ACC	AUC
PFA-LR	0.798	0.840[3]
PFA-XGBOOST	**0.799**	**0.882**
IRT-LR	0.798	0.863[3]
IRT-XGBOOST	**0.798**	**0.867**

Sources: [1] (Yeung & Yeung, 2018),[2] (Gervet et al., 2020),[3] (Arkiza et al., 2023).

comparable to those reported in prior research on specific datasets. These studies have used the same number of trees and regularization parameters, but without a learning rate, which negatively impacts results by reducing the AUC value. Given that the boosting process involves 1000 trees with a learning rate of 0.1 for each iteration, the convergence speed and computational complexity of each iteration can affect the overall training time. This ultimately influences the model's performance.

IRT is a simple model that focuses primarily on feature difficulty. Due to the complexity of this feature, it often requires more iterations to capture its patterns effectively, which can increase training time, particularly on large datasets. This observation supports the consistency between our results and established references, especially when working with smaller datasets such as ***assist15***, ***assist17***, statics, and Moodle_Morocco. In these cases, the results of logistic regression and XGBoost are quite similar.

However, the model's predictive power diminishes with larger datasets like ***assist12***, ***algebra05***, ***bridge_algebra06***, and ***assist09***, where the AUC tends to decrease (see Table 7). Consequently, logistic regression tends to yield better results for these larger datasets. Despite XGBoost's robustness, it can be challenging to interpret complex interactions within IRT, where responses to questions and latent traits may not be easily captured by a tree-based model. In contrast, logistic regression, with its linear assumptions, tends to handle these situations better.

PFA is generally more practical and realistic compared to IRT. One reason is that IRT treats the learner's knowledge as static, assuming that a learner's ability remains unchanged after completing a skill. In contrast, PFA takes into account a learner's history, tracking both successful and unsuccessful attempts in skill-based exercises. This approach allows for a more dynamic view of learner performance. Additionally, PFA's method for predicting performance is mathematically straightforward, as it relies on counting historical attempts to gauge skill-based progress.

Our evaluation of the XGBoost model in DAS3H has provided significant insights into predicting learner performance. Through comprehensive analysis and testing, the model delivered outstanding results on the assist17 dataset, with an AUC improvement from 0.690 to 0.709. This increase of +0.019 represents a notable enhancement compared to previous work and equals the progress achieved by the BestLR model on assist17. On other datasets from ASSISTments and KDD that contain temporal characteristics, the model's results remained relatively consistent, demonstrating the robustness of both models XGBoost and logistic regression, and their ability to capture similar relationships and patterns across various datasets. This similarity can be attributed to common preprocessing and encoding stages, which allowed both logistic regression and XGBoost to produce nearly identical AUC and accuracy scores. This consistency underscores the reliability and generalizability of these models across different datasets.

DAS3H has shown a significant improvement in prediction performance compared to IRT and PFA when used with both XGBoost and Logistic Regression. The key difference in DAS3H is its ability to use different parameters for different skills and incorporate time-window characteristics instead of simple mathematical counts of correct attempts. This resulted in DAS3H achieving the highest AUC and ACC. According to an ablation study by (Gervet et al., 2020), the performance boost in DAS3H compared to PFA primarily comes from the addition of item difficulty, rather than the use of temporal windows. Given this finding, we decided not to include multidimensional embedding and pairwise interactions, as previous work indicated they did not improve model performance. In fact, setting the value to "d = 20″ led to the worst evaluation metrics, as reported by (Choffin et al., 2019).

Calibration designates the consistency between the predicted probabilities generated within a model and the real observed results. According to a well-calibrated model, the predicted probabilities reflect the true probability of an event. The XGBoost and Logistic Regression in the context of DAS3H were subjected to reliability analysis, through calibration plots of the mean predicted probabilities against the fraction of positive, to detect biases and correct them before spreading into learner performance application (Gervet et al., 2020). Therefore, the calibration analysis did not reveal any biases, and the models have achieved a perfect calibration across diverse datasets - ***assist17***, ***algebra05***, ***bridge06***, ***assist12*** – as illustrated in Fig. 3.

7.1. Implications of our findings

Our findings suggest that XGBoost's boosting technique offers significant advantages over traditional Logistic Regression, particularly in handling complex datasets and improving predictive accuracy. This is crucial for applications such as learner performance prediction, where accurate and reliable predictions can drive better educational outcomes. The enhancements in AUC across multiple datasets highlight XGBoost's robustness and its potential as a valuable tool in educational data mining.

In the context of IRT, our XGBoost model achieved performance levels comparable to prior research but with better AUC values due to the inclusion of a learning rate. This underscores the importance of considering learning rates in boosting algorithms to improve convergence speed and overall model performance. However, the increased training time and computational complexity, especially for large datasets, suggest a trade-off that needs careful consideration.

For PFA, XGBoost's dynamic approach in tracking learners' histories

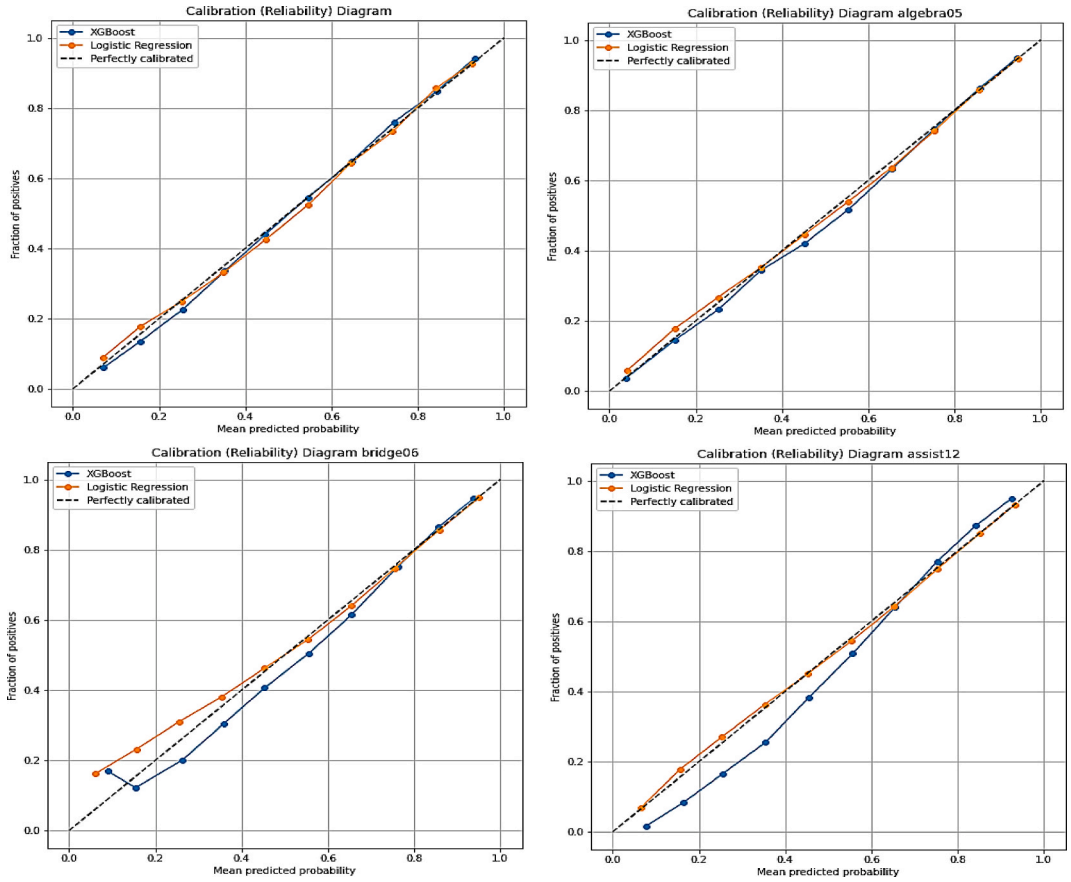

Fig. 3. LR VS XGBoost calibration diagrams for assist17, algebra05, bridge06, and assist12 (DAS3H).

provided a more practical and realistic prediction of performance compared to the static nature of IRT. The consistent performance of XGBoost across different datasets further supports its reliability and generalizability.

7.2. Future research

Future research should explore the integration of multidimensional embeddings and pairwise interactions in the boosting framework, despite previous indications of limited performance improvements. These aspects could potentially uncover deeper insights into learner behaviors and interactions. Additionally, examining the impact of different regularization techniques and hyperparameter tuning strategies on XGBoost's performance could provide further optimization avenues.

Further studies should also investigate the scalability of XGBoost in real-time applications, particularly in educational settings where timely predictions are critical. Exploring hybrid models that combine the strengths of both Logistic Regression and XGBoost might yield even more robust predictive models.

The results of our study underscore the potential of boosting techniques, particularly XGBoost, to serve as a powerful alternative to traditional Logistic Regression models. The significant improvements in predictive accuracy and robustness across various datasets suggest that educational institutions and researchers can rely on XGBoost for more accurate learner performance predictions. This can facilitate better decision-making and personalized learning interventions, ultimately enhancing educational outcomes.

The study also highlights the importance of hyperparameter optimization and the inclusion of learning rates in boosting algorithms. These factors play a crucial role in enhancing model performance and should be considered in future applications and research.

Moving forward, research should focus on addressing the limitations identified in our study. This includes investigating methods to reduce the computational complexity and training time associated with XGBoost, especially for large datasets. Future work could also explore the integration of advanced features such as multidimensional embeddings and pairwise interactions to further improve predictive performance. Moreover, there is a need for more comprehensive studies on the application of XGBoost in different educational contexts and datasets. Such studies could provide deeper insights into the model's adaptability and effectiveness in various scenarios, contributing to the broader field of educational data mining. Additionally, exploring the combination of XGBoost with other machine learning techniques could lead to the development of more robust and accurate predictive models.

8. Conclusion

Predicting learner performance has garnered significant interest among researchers due to its critical role in enhancing educational quality and the effectiveness of intelligent tutoring systems. It helps in adapting these systems to individual learners' needs. This process involves analyzing historical ITS data to estimate the likelihood of students answering subsequent questions correctly.

In this study, we aim to improve learner performance prediction. We explore how logistic regression-based models such as Item Response Theory, Performance Factor Analysis, and DAS3H could be enhanced by one of the most accurate machine learning models, known for its ability to solve various data science problems: the XGBoost model. XGBoost has demonstrated superior predictive performance, outperforming state-of-the-art results in seven datasets for the PFA model, achieving significant AUC values with appropriate hyperparameter tuning. Additionally,

XGBoost has improved the performance prediction of the DAS3H model on the ASSISTment17 dataset, where the AUC increased from 0.690 to 0.709. For the IRT model, XGBoost maintained consistent AUC results across multiple datasets.

Although the XGBoost model is known for its strength, scalability, and impressive results, it requires further refinement to excel in all logistic regression-based models, particularly IRT and DAS3H. In these cases, logistic regression sometimes produces better results. To address this, additional work is needed, especially in the areas of hyperparameter tuning to optimize performance. Additionally, special encoding of specific learner features for XGBoost could unlock its potential, leading to more accurate results. A significant limitation of XGBoost is its execution time, which can be prohibitively long for large datasets. To overcome this, future research should focus on optimizing execution times and exploring different approaches to improve the accuracy of student performance predictions. By addressing these issues, researchers could maximize XGBoost's potential in educational contexts.

Statements on open data and ethics

The ASSISTments, KDD, and Statics datasets used in this study are publicly available on GitHub, with the reference: (Theo, 2020). The Moodle_Morocco dataset can be requested from the corresponding author. Additional information on this dataset is provided in the related article (Arkiza et al., 2023).

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CRediT authorship contribution statement

Soukaina Hakkal: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ayoub Ait Lahcen:** Validation, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

List of acronyms

IRT	Item Response Theory
PFA	Performance Factor Analysis
DAS3H	Dynamic Assessment System for Holistic Heuristic
ITS	Intelligent Tutoring System
XGBoost	eXtreme Gradient Boosting

References

Alsulami, A. A., AL-Ghamdi, A. S. A.-M., & Ragab, M. (2023). Enhancement of E-learning student's performance based on ensemble techniques. *Electronics, 12*(6), 1508. https://doi.org/10.3390/electronics12061508

An, X., & Yung, Y.-F. (2014). Item response theory: What it is and how you can use the IRT procedure to apply it. *SAS Institute Inc, 10*(4). *SAS364-2014*.

Anderson, J. R., Boyle, C. F., & Reiser, B. J. (1985). Intelligent tutoring systems. *Science, 228*(4698), 456–462. https://doi.org/10.1126/science.228.4698.456

Arkiza, M., Hakkal, S., Oumaira, I., & Ait Lahcen, A. (2023). A comparative study of adaptative learning algorithms for students' performance prediction: Application in a Moroccan university computer science course. In *International conference on advanced intelligent systems for sustainable development*. https://doi.org/10.1007/978-3-031-26384-2_61, 698–709.

Asselman, A., Khaldi, M., & Aammou, S. (2023). Enhancing the prediction of student performance based on the machine learning XGBoost algorithm. *Interactive Learning Environments, 31*(6), 3360–3379. https://doi.org/10.1080/10494820.2021.1928235

Atkinson, R. C., & Wilson, H. A. (1968). Computer-assisted instruction. *Science, 162*(3849), 73–77. https://doi.org/10.1126/science.162.3849.73

Cen, H., Koedinger, K., & Junker, B. (2006). Learning factors analysis - a general method for cognitive model evaluation and improvement. In M. Ikeda, K. D. Ashley, & T.-W. Chan (Eds.), *Proc. The 8th international conference on intelligent tutoring systems* (pp. 164–175). Springer Berlin Heidelberg. https://doi.org/10.1007/11774303.

Chen, T., & Guestrin, C. (2016). XGBoost. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. https://doi.org/10.1145/2939672.2939785

Chen, Y., & Zhai, L. (2023). A comparative study on student performance prediction using machine learning. *Education and Information Technologies, 28*(9), 12039–12057. https://doi.org/10.1007/s10639-023-11672-1

Choffin, B., Popineau, F., Bourda, Y., & Vie, J.-J. (2019). DAS3H: Modeling student learning and forgetting for optimally scheduling distributed practice of skills. In *Proceedings of the 12th international conference on educational data mining (EDM 2019)* (pp. 29–33).

Corbett, A. T., & Anderson, J. R. (1995). Knowledge tracing: Modeling the acquisition of procedural knowledge. *User Modeling and User-Adapted Interaction, 4*(4), 253–278. https://doi.org/10.1007/BF01099821

Ekanadham, C., & Karklin, Y. (2017). *T-SKIRT: Online estimation of student proficiency in an adaptive learning system*.

Fan, Z., Gou, J., & Wang, C. (2023). Predicting secondary school student performance using a double particle swarm optimization-based categorical boosting model. *Engineering Applications of Artificial Intelligence, 124*, Article 106649. https://doi.org/10.1016/j.engappai.2023.106649

Feng, M., Heffernan, N., & Koedinger, K. (2009). Addressing the assessment challenge with an online system that tutors as it assesses. *User Modeling and User-Adapted Interaction, 19*(3), 243–266. https://doi.org/10.1007/s11257-009-9063-7

Freund, Y., Schapire, R., & Abe, N. (1999). A short introduction to boosting. *Journal of Japanese Society for Artificial Intelligence, 14*(771–780), 1612.

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics, 29*(5), 1189–1232. http://www.jstor.org/stable/2699986.

Gervet, T., Koedinger, K., Schneider, J., & Mitchell, T. (2020). When is deep learning the best approach to knowledge tracing? *Journal of Educational Data Mining, 12*(3), 31–54.

Gong, Y., Beck, J. E., & Heffernan, N. T. (2010). Comparing knowledge tracing and performance factor analysis by using multiple model fitting procedures. ITS 2010. In *Intelligent tutoring systems, 6094* pp. 35–44). https://doi.org/10.1007/978-3-642-13388-6_8. Springer, Berlin, Heidelberg.

Hakkal, S., & Ait Lahcen, A. (2021). An overview of adaptive learning fee-based platforms. In *Proceedings of the 2nd international conference on Big data, modelling and machine learning*. https://doi.org/10.5220/0010731400003101, 222–226.

Hsu, C.-K., Hwang, G.-J., & Chang, C.-K. (2010). Development of a reading material recommendation system based on a knowledge engineering approach. *Computers & Education, 55*(1), 76–83. https://doi.org/10.1016/j.compedu.2009.12.004

Kass, R. (1989). Student modeling in intelligent tutoring systems — implications for user modeling. In *User models in dialog systems* (pp. 386–410). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-83230-7_14.

Ke, G., Qi, M., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems, 30*.

Koedinger, K. R., Baker, R. S., Cunningham, K., Skogsholm, A., Leber, B., & Stamper, J. (2010). A data repository for the EDM community: The PSLC DataShop. In *Handbook of educational data mining, 43* pp. 43–56).

Kosmas, P., & Zaphiris, P. (2023). Improving students' learning performance through technology-enhanced embodied learning: A four-year investigation in classrooms. *Education and Information Technologies, 28*(9), 11051–11074. https://doi.org/10.1007/S10639-022-11466-X/METRICS

Lai, Z., Wang, L., & Ling, Q. (2021). Recurrent knowledge tracing machine based on the knowledge state of students. *Expert Systems, 38*(8). https://doi.org/10.1111/exsy.12782

Lindsey, R. V., Shroyer, J. D., Pashler, H., & Mozer, M. C. (2014). Improving students' long-term knowledge retention through personalized review. *Psychological Science, 25*(3), 639–647. https://doi.org/10.1177/0956797613504302

Liu, T. (2022). Knowledge tracing: A bibliometric analysis. *Computers in Education: Artificial Intelligence, 3*, Article 100090. https://doi.org/10.1016/J.CAEAI.2022.100090

Matzavela, V., & Alepis, E. (2021). Decision tree learning through a predictive model for student academic performance in intelligent M-learning environments. *Computers in Education: Artificial Intelligence, 2*, Article 100035. https://doi.org/10.1016/j.caeai.2021.100035

Nagatani, K., Zhang, Q., Sato, M., Chen, Y.-Y., Chen, F., & Ohkuma, T. (2019). Augmenting knowledge tracing by considering forgetting behavior. *The World Wide Web Conference*, 3101–3107. https://doi.org/10.1145/3308558.3313565

Nahar, K., Shova, B. I., Ria, T., Rashid, H. B., & Islam, A. H. M. S. (2021). Mining educational data to predict students performance. *Education and Information Technologies, 26*(5), 6051–6067. https://doi.org/10.1007/s10639-021-10575-3

Nwana, H. S. (1990). Intelligent tutoring systems: An overview. *Artificial Intelligence Review, 4*(4), 251–277. https://doi.org/10.1007/BF00168958

Ojajuni, O., Ayeni, F., Akodu, O., Ekanoye, F., Adewole, S., Ayo, T., Misra, S., & Mbarika, V. (2021). Predicting student academic performance using machine

learning. *Computational Science and Its Applications – ICCSA 2021, 12957*, 481–491. https://doi.org/10.1007/978-3-030-87013-3_36. Springer, Cham.

Pavlik Jr, P. I., Cen, H., & Koedinger, K. R. (2009). Performance factors analysis-A new alternative to knowledge tracing. In *Proceeding of the 14th international conference on artificial intelligence in education*.

Piech, C., Spencer, J., Huang, J., Ganguli, S., Sahami, M., Guibas, L., & Sohl-Dickstein, J. (2015). *Deep knowledge tracing*.

Reckase, M. D. (2009). Multidimensional item response theory models. In *Statistics for social and behavioral sciences multidimensional item response theory* (pp. 79–112). Springer. http://www.springer.com/3463.

Rodríguez-Hernández, C. F., Musso, M., Kyndt, E., & Cascallar, E. (2021). Artificial neural networks in academic performance prediction: Systematic implementation and predictor evaluation. *Computers in Education: Artificial Intelligence, 2*, Article 100018. https://doi.org/10.1016/j.caeai.2021.100018

Schmucker, R., Wang, J., Hu, S., & Mitchell, T. M. (2021). *Assessing the performance of online students – new data.* New Approaches, Improved Accuracy.

Sein, M. (2022). AI-assisted knowledge assessment techniques for adaptive learning environments. *Computers in Education: Artificial Intelligence, 3*. https://doi.org/10.1016/j.caeai.2022.100050

Shute, V. J. (2008). Focus on formative feedback. *Review of Educational Research, 78*(1), 153–189. https://doi.org/10.3102/0034654307313795

Stamper, J., Niculescu-Mizil, A., Ritter, S., Gordon, G. J., & Koedinger, K. R. (2010). *Algebra I 2005-2006 and Bridge to Algebra 2006-2007. Development data set from KDD Cup 2010 Educational Data Mining Challenge*. http://pslcdatashop.web.cmu.edu/KDDCup/downloads.jsp.

Taghizadeh Kerman, N., Kazem Banihashem, S., Karami, M., Er, E., van Ginkel, S., & Noroozi, O. (2024). Online peer feedback in higher education: A synthesis of the literature. *Education and Information Technologies, 29*, 763–813. https://doi.org/10.1007/s10639-023-12273-8

Theo. (2020). Learner performance prediction. *Github*. https://github.com/theophilegervet/learner-performance-prediction.

van der Linden, W. J., & Hambleton, R. K. (2013). *Handbook of item response theory*.

Vie, J.-J., & Kashima, H. (2019). Knowledge tracing machines: Factorization machines for knowledge tracing. *Proceedings of the AAAI Conference on Artificial Intelligence, 33* (1), 750–757. https://doi.org/10.1609/aaai.v33i01.3301750

Wang, T. H. (2007). What strategies are effective for formative assessment in an e-learning environment? *Journal of Computer Assisted Learning, 23*(3), 171–186. https://doi.org/10.1111/j.1365-2729.2006.00211.x

Wilson, M., & De Boeck, P. (2004). Descriptive and explanatory item response models. In *Explanatory item response models* (pp. 43–74). New York: Springer. https://doi.org/10.1007/978-1-4757-3990-9_2.

Xiong, X., Zhao, S., Van Inwegen, E. G., & Beck, J. E. (2016). Going deeper with deep knowledge tracing. In *Proceedings of the 9th international conference on educational data mining* (pp. 545–550).

Yan, K. (2021). Student performance prediction using XGBoost method from A macro perspective. In *2021 2nd international conference on computing and data science (CDS)*. https://doi.org/10.1109/CDS52072.2021.00084, 453–459.

Yeung, C.-K., & Yeung, D.-Y. (2018). Addressing two problems in deep knowledge tracing via prediction-consistent regularization. In *Proceedings of the fifth annual ACM conference on learning at scale*. https://doi.org/10.1145/3231644.3231647, 1–10.